

Question 3-58

$$\dot{Q} = 75 \text{ W}, A = 0.1 \text{ m}^2, h = 10 \text{ W/m}^2 \cdot \text{K}, T_1 = 20^\circ \text{C}, T_2 = ?$$

$$\dot{Q} = h A \Delta T$$

$$\Delta T = \frac{\dot{Q}}{h A}$$

$$(T_2 - T_1) = \frac{\dot{Q}}{h A}$$

$$T_2 = \frac{(75)}{(0.1)(10)} + 20 = 95^\circ \text{C}$$

Question 3-59

$$A = 3 \text{ m}^2, T_1 = 75^\circ \text{C}, T_2 = 18^\circ \text{C}, k = 0.08$$

$$\dot{Q} = 200 \text{ W}$$

$$\dot{Q}_{\text{cond}} = k A \frac{\Delta T}{\Delta x} \quad \Delta x = \text{distance}$$

$$\Delta x = \frac{(0.08)(3)(75-18)}{200} = 0.068 \text{ m}$$

Question 3-60

$$\Delta x = 1.5 \text{ cm} = 1.5 \times 10^{-2} \text{ m}, k = 0.16, \Delta T = 20^\circ \text{C}$$

$$\dot{Q} = \frac{Q}{A} = k \frac{\Delta T}{\Delta x}$$

$$k \frac{\Delta T}{\Delta x} = (0.16) \frac{(20)}{0.015} = 213.33 \text{ W/m}^2$$

$$h = h_1 + u = 2487.2 \text{ kJ/kg} \quad \therefore h_1 = h_2$$

Now for quality at $P = 10 \text{ kPa}$

$$h = h_f + x h_{fg}$$

$$x = \frac{h - h_f}{h_{fg}} = \frac{2487.2 - 191.81}{2392.82} = 0.95$$

and at $P = 10 \text{ kPa}, T = 45.8^\circ \text{C}$

Question 4-49

$$\dot{m} = 0.05 \text{ kg/s}, P_1 = 1000 \text{ kPa}, T_1 = 550 \text{ K}$$

$$P_2 = 250 \text{ kPa} \text{ and } T_2 = 300 \text{ K}, W = 55 \text{ kW}$$

$$\dot{Q} = ?$$

As we know,

$$\dot{m} h_1 + \dot{Q} = \dot{m} h_2 + \dot{W}$$

$$\dot{Q} = \dot{m} (h_2 - h_1) + \dot{W}$$

As we don't have values of h , we'll use

$$\dot{Q} = \dot{m} c_p (T_2 - T_1) + \dot{W}$$

$$\dot{Q} = (0.05)(519.3)(300 - 550) + 55$$

$$\dot{Q} = -9.9 \text{ kW}$$

$$= 415.5 \text{ kJ/kg}$$

Put this in eq ①

$$\dot{m} = \frac{5000}{415.5} = 12 \text{ kg/s}$$

Question 4-126

$$T_1 = 20^\circ \text{C}, P_1 = 200 \text{ kPa}, \dot{Q} = 0$$

$$T_2 = ?$$

We have a transient problem here

$$m_2 - 0 = m_1$$

$$m_2 u_2 - 0 = m h_1 + \dot{Q} - W = m h_1 + 0 - 0$$

$$u_2 = h_1 \text{ and } P_2 = P_1$$

This shows that $T_2 > T_1$ and Temp do not depend on volume.

Question 4-132

$$T_1 = 25^\circ \text{C}, P_1 = 8 \text{ MPa}, P_2 = 6 \text{ MPa}, P_3 = ?$$

Now we know, $\dot{m}_1 = \dot{m}_2$

$$\dot{m}_1 h_1 = \dot{m}_2 h_2 \text{ where } u_2 = h_1$$

$$T_2 = (c_p / c_v) T_1 = k T_1 = 1.4 (298) = 417.5 \text{ K}$$

$$P_3 = P_2 \times \frac{T_3}{T_2} = 6 \text{ MPa} \times \frac{298.15}{417.5} = 4.29 \text{ MPa}$$

4.24 In a jet engine a flow of air at 1000 K, 200 kPa, and 40 m/s enters a nozzle, where the air exits at 500 m/s, 90 kPa. What is the exit temperature, assuming no heat loss?

4.25 Superheated vapor ammonia enters an insulated nozzle at 30°C , 1000 kPa, as shown in Fig. P4.25, with a low velocity and at a rate of 0.01 kg/s. The ammonia exits at 300 kPa with a velocity of 450 m/s. Determine the temperature (or quality, if saturated) and the exit area of the nozzle.

4.54 A factory generates compressed air from 100 kPa, 17°C by compression to 1000 kPa, 600 K, after which it cools in a constant pressure cooler to 300 K, (see Fig. P4.54). Find the specific compressor work and the specific heat transfer in the cooler.

Question 3-61

$$A = 2 \text{ m}^2, T_1 = 15^\circ \text{C}, T_2 = 2^\circ \text{C}, h = 125 \text{ W/m}^2 \cdot \text{K}$$

$$\dot{Q} = ?$$

$$\dot{Q} = h A \Delta T$$

$$\dot{Q} = (125)(2)(15-2) = 3250 \text{ W}$$

Question 3-62

$$Q_1 = 25 \text{ W}, Q_2 = 50 \text{ W}, T_1 = 20^\circ \text{C}, A = 1 \text{ m}^2$$

$$K = 15 \text{ W/m}^2 \cdot \text{K}$$

$$Q_{\text{tot}} = 25 + 50 = 75 \text{ W}$$

$$\dot{Q} = \frac{\Delta T}{\frac{A}{hA}} = \frac{75}{(15)(1)} = 5^\circ \text{C}$$

$$5 + 20 = 25^\circ \text{C} \text{ (required)}$$

Question 4-23

$$T_1 = 1000 \text{ K}, P_1 = 200 \text{ kPa}, V_1 = 30 \text{ m/s}$$

$$T_2 = 850 \text{ K}, P_2 = 90 \text{ kPa}, V_2 = ?$$

As

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

and

$$\dot{m} (h_1 + \frac{V_1^2}{2}) = \dot{m} (h_2 + \frac{V_2^2}{2})$$

Now,

$$\frac{h_1 + \frac{V_1^2}{2}}{2} = \frac{h_2 + \frac{V_2^2}{2}}{2} \quad \therefore m \text{ got cancelled because of same value}$$

Question no 4-50

$$T_1 = 25^\circ \text{C}, k = A, P_1 = 1000 \text{ kPa}, T_2 = 40^\circ \text{C}$$

No kinetic energy

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{m} h_1 = \dot{m} h_2 + \dot{W}$$

$$\dot{W} = \dot{m} (h_1 - h_2) = (2108)(40 - 25) = 31620 \text{ kJ/kg}$$

$$h_2 = 316.05 \text{ kJ/kg} \text{ (at } P_2 = 1000 \text{ kPa, } T_2 = 40^\circ \text{C)}$$

$$\dot{W} = \dot{m} (h_1 - h_2) = (259.77) - (316.05) = -46.28 \text{ kJ/kg}$$

Question no 4-54

$$T_1 = 17^\circ \text{C} \text{ and } P_1 = 100 \text{ kPa}, T_2 = 600 \text{ K} \text{ and } P_2 = 1 \text{ MPa}$$

$$T_3 = 300 \text{ K}, W_c = P, \dot{Q} = ?$$

As we know,

$$\dot{m}_1 = \dot{m}_2 = \dot{m}_3 \text{ when } \dot{Q} = 0$$

$$\dot{Q} = \dot{m}_1 h_1 + \dot{W}_1 - \dot{m}_2 h_2 = 0$$

We will find h_1 and h_2 from table

$$\dot{m}_1 h_1 = \dot{m}_2 h_2$$

$$T_1 = 17 + 273 = 290 \text{ K}, h_1 = 290.19 \text{ kJ/kg}$$

$$T_2 = 600 \text{ K}, h_2 = 607.32 \text{ kJ/kg}$$

$$T_3 = 300 \text{ K}, h_3 = 300.47 \text{ kJ/kg}$$

$$\dot{W}_{in} = \dot{m}_2 h_2 - \dot{m}_1 h_1 = (607.32) - (290.19) = 317.13 \text{ kJ/kg}$$

Question 4-133

$$V = 2 \text{ m}^3, P_1 = 3 \text{ MPa}, T_1 = 90^\circ \text{C}, P_2 = 3 \text{ MPa}$$

$$\text{mass} = P \text{ and } T_2 = P$$

Now

$$\text{we know } m_2 - m_1 = m_{in}$$

$$m_2 u_2 - m_1 u_1 = -m_{in} h_{in} + \dot{Q}_2 - \dot{W}_2$$

$$\dot{Q} = 0, V = 0, W = 0$$

$$m_1 = 0, m_2 = m_{in}$$

$$P_2 = P_1 = 3 \text{ MPa}$$

$$u_2 = h_{in} = 436.19 \text{ kJ/kg} \text{ (from table)}$$

Now by interpolation

$$T_2 = 100 + 10 \frac{436.19 - 433.77}{446.48 - 433.77} = 101.9^\circ \text{C}$$

$$V_2 = 0.00665 + 0.1904 (0.00734 - 0.00665) = 0.006781 \text{ m}^3/\text{kg}$$

Now

$$m = V / V_2 = (2 / 0.006781) = 294.9 \text{ kg}$$

4.55 A compressor brings R-134a from 150 kPa, -10°C to 1200 kPa, 50°C . It is water cooled, with heat loss estimated as 40 kW, and the shaft work input is measured to be 150 kW. What is the mass flow rate through the compressor?

4.56 The compressor of a large gas turbine receives air from the ambient surroundings at 95 kPa, 20°C with low velocity. At the compressor discharge, air exits at 1.52 MPa, 430°C with a velocity of 90 m/s. The power input to the compressor is 5000 kW. Determine the mass flow rate of air through the unit.

4.23 In a jet engine a flow of air at 1000 K, 200 kPa, and 30 m/s enters a nozzle, as shown in Fig. P4.23, where the air exits at 850 K, 90 kPa. What is the exit velocity, assuming no heat loss?

h will come from Table A-7.1 (air)

$$h_1 = 1046.22, h_2 = 877.40$$

$$h_1 + \frac{V_1^2}{2} - h_2 = \frac{V_2^2}{2}$$

$$\frac{1046.22 + (30^2)}{2000} - \frac{877.40}{2000} = \frac{V_2^2}{2000}$$

$$V_2 = 531.8 \text{ m/s}$$

Question 4-24

$$T = 1000 \text{ K}, P_1 = 200 \text{ kPa}, V_1 = 40 \text{ m/s}$$

$$V_2 = 500 \text{ m/s}, P_2 = 90 \text{ kPa}, T_2 = ?$$

Now

$$h_1 + \frac{V_1^2}{2} - \frac{V_2^2}{2} = h_2$$

$$\frac{1046.22 + (40^2)}{2000} - \frac{(500^2)}{2000} = h_2 = 922.02 \text{ kJ/kg}$$

For T_2 we will do interpolation

As $h_2 = 922.02$ we will take 2 values close to it from table A-7.1

T	h
900	933.15
x	922.02
850	877.40

$$\frac{900 - x}{933.15 - 922.02} = \frac{900 - 850}{933.15 - 877.40}$$

$$x = 890.01$$

$$\therefore T_2 = 890 \text{ K}$$

$$W = h_2 - q_{\text{out}} - h_3$$

$$0 = h_2 - q_{\text{out}} - h_3$$

$$q_{\text{out}} = h_2 - h_3$$

$$q_{\text{out}} = 607.32 - 300.47 = 306.85 \text{ kJ/kg}$$

Qno 4-55

$$P_1 = 150 \text{ kPa}, T_1 = -10^\circ \text{C}, P_2 = 1200 \text{ kPa}, T_2 = 50^\circ \text{C}$$

First we will find h

$$h_1 = 393.84 \text{ kJ/kg}, h_2 = 426.84 \text{ kJ/kg}$$

Now

$$\dot{m} h_1 + \dot{Q} = \dot{m} h_2 + \dot{W}$$

As $\dot{m}_1 = \dot{m}_2$

$$\dot{Q} = 40 \text{ kW}, \dot{W} = 150 \text{ kW}$$

$$\dot{m} = \frac{(\dot{Q} - \dot{W})}{h_2 - h_1} = \frac{(40 - 150)}{426.84 - 393.84} = 3.33 \text{ kg/s}$$

Question 4-56

$$P_1 = 95 \text{ kPa}, T_1 = 20^\circ \text{C}, V_1 = 0, P_2 = 1.52 \text{ MPa}$$

$$T_2 = 430^\circ \text{C}, V_2 = 90 \text{ m/s}, W_c = 5000 \text{ W}, \dot{m} = ?$$

As we know

$$\dot{m} = \frac{W_c}{-W}$$

So find W , we have $q + h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} + W$

$$q = 0, V_1 = 0$$

$$-W = c_p (T_2 - T_1) + \frac{V_2^2}{2} = (1.004)(430 - 20) + \frac{(90^2)}{2000}$$

3.58 The brake shoe and steel drum of a car continuously absorb 75 W as the car slows down. Assume a total outside surface area of 0.1 m^2 with a convective heat transfer coefficient of $10 \text{ W/m}^2 \cdot \text{K}$ to the air at 20°C . How hot does the outside brake and drum surface become when steady conditions are reached?

3.59 A water heater is covered with insulation boards over a total surface area of 3 m^2 . The inside board surface is at 75°C , the outside surface is at 18°C , and the board material has a conductivity of $0.08 \text{ W/m} \cdot \text{K}$. How thick should the board be to limit the heat transfer loss to 200 W ?

3.60 Find the rate of conduction heat transfer through a 1.5-cm -thick hardwood board, $k = 0.16 \text{ W/m} \cdot \text{K}$, with a temperature difference between the two sides of 20°C .

3.61 A 2-m^2 window has a surface temperature of 15°C , and the outside wind is blowing air at 2°C across it with a convection heat transfer coefficient of $h = 125 \text{ W/m}^2 \cdot \text{K}$. What is the total heat transfer loss?

3.62 Owing to a faulty door contact, the small light bulb (25 W) inside a refrigerator is kept on and limited insulation lets 50 W of energy from the outside seep into the refrigerated space. How much of a temperature difference from the ambient surroundings at 20°C must the refrigerator have in its heat exchanger with an area of 1 m^2 and an average heat transfer coefficient of $15 \text{ W/m}^2 \cdot \text{K}$ to reject the leaks

4.49 A small expander (a turbine with heat transfer) has 0.05 kg/s helium entering at 1000 kPa , 550 K and leaving at 250 kPa , 300 K . The power output on the shaft measures 55 kW . Find the rate of heat transfer, neglecting kinetic energies.

4.48 A small turbine, shown in Fig. P4.48, is operated at part load by throttling a 0.25-kg/s steam supply at 1.4 MPa and 250°C down to 1.1 MPa before it enters the turbine, and the exhaust is at 10 kPa . If the turbine produces 110 kW , find the exhaust temperature (and quality if saturated).

- 4.132** An evacuated 150-L tank is connected to a line flowing air at room temperature, 25°C , and 8 MPa pressure. The valve is opened, allowing air to flow into the tank until the pressure inside is 6 MPa. At this point the valve is closed. This filling process occurs rapidly and is essentially adiabatic. The tank is then placed in storage, where it eventually returns to room temperature. What is the final pressure?
- 4.133** An insulated 2-m^3 tank is to be charged with R-134a from a line flowing the refrigerant at 3 MPa, 90°C . The tank is initially evacuated, and the valve is closed when the pressure inside the tank reaches 3 MPa. Find the mass in the tank and its final temperature.
- 4.126** An initially empty cylinder is filled with air from 20°C , 100 kPa until it is full. Assuming no heat transfer, is the final temperature above, equal to, or below 20°C ? Does the final T depend on the size of the cylinder?